

Networked Optical Massive MIMO Communications

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Abstract—The low cost and versatility of optical devices make it possible to pack a large number of optical transceivers into arrays and exploit massive multiple-input multiple-output (MIMO) transmission in optical wireless communications. Nevertheless, optical massive MIMO presents several distinct challenges such as, the line-of-sight propagation and intensity modulation, incompatible with existing radio frequency massive MIMO techniques. This paper presents a networked optical massive MIMO system that consists of multiple base stations (BSs), each equipped with a transmit lens and an optical transmitter array, cooperatively serving a number of user terminals (UTs), each equipped with a receive lens and a photodetector array. We establish the optical massive MIMO channel model, analyze its asymptotic behavior, and evaluate the potential of networked optical massive MIMO on system throughput improvement. To achieve high throughput, we propose optical beam division multiple access (BDMA) transmission schemes under the total and per transmitter power constraints with the asymptotic optimality. Our results show that the system sum rate increases proportionally to the number of BSs and UTs using the optical BDMA transmission. Further numerical results show that the proposed optical massive MIMO system along with the optical BDMA transmission is able to achieve high throughput with low complexity.

Index Terms—Networked optical massive MIMO communication, transmit and receive lenses, beam division multiple access (BDMA) transmission.

I. INTRODUCTION

THE rapidly rising demand for high data rate compounded by the problem of spectrum shortage in radio

frequency (RF) communications, motivates many novel technologies using higher frequency spectrum resources. One of the promising alternative technologies is optical wireless communication [1], where base stations (BSs) employ optical transmitters such as light-emitting diodes (LEDs) [2] or laser diodes (LDs) [3] to convert the electrical signals to the optical signals, and user terminals (UTs) employ photodetectors to convert the light intensity to the corresponding electrical signals. Alternatively, optical fiber ports can be used to transmit and receive optical signals [4]. Many studies have demonstrated that optical wireless communication has the potential to achieve high transmission capacity with low-cost devices, which makes it a promising technology for various future applications [5]–[7].

To further increase the transmission rate, the use of multiple-input multiple-output (MIMO) technique is a natural progression. Specifically, multi-user multiple-input single-output (MU-MISO) technique comes out first to support multi-user optical wireless communications [8], [9], where several separate BSs (e.g., LED arrays) are required to provide spatial multiplexing. To mitigate the inter-user interference, linear precoding schemes minimizing the mean squared error (MSE) [9] or maximizing the signal-to-interference-plus-noise ratio (SINR) [8] have been studied. Employing single photodetector at the UT side may lead to coverage problems and degradation in system performance. Hence, UTs can employ a imaging receiver with a receive lens and multiple photodetectors [10] to separate signals from different directions, potentially offering higher spatial diversity [11], [12]. Based on imaging receivers, a series of works have studied the performance of precoding schemes in MU-MIMO optical communication systems [8], [13]–[16]. The authors in [8] compared the performance of the zero forcing and dirty paper coding schemes. The work [13] derived an optimal linear precoding transmitter based on the minimum mean-squared error (MMSE) criterion, and the block diagonalization precoding algorithm was investigated in [14]. Several works investigated the precoding designs based on sum rate maximization [15] and max-min fairness [16]. Furthermore, recently proposed massive MIMO with tens or hundreds of antennas in RF systems [17] was applied into optical communication systems [18] to increase spectrum efficiency.

In the above works, BSs and UTs are equipped with a number of LEDs and photodetectors, respectively. However, as optical wireless communication systems employ intensity modulation with direct detection (IM/DD) [5] and usually

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work in line-of-sight (LOS) scenarios, the elements of channel matrix are highly correlated. Specifically, because the distance between the BS and one UT is much larger than the LED array size, the optical channel gains from different LEDs to one photodetector at the UT are highly correlated [10]. Even the imaging receiver with a receive lens cannot separate the signals from different LEDs in one LED array. Consequently, the optical MIMO channel matrix often exhibits the rank-one property, implying that the multiple LEDs in one BS only play as one transmit antenna and a BS can only transmit a single data stream to each UT.

To break through the restrict of the number of served UTs, we have proposed beam domain optical wireless massive MISO communications with a transmit lens [19]. By employing the transmit lens, lights from different transmitters are refracted to different directions with sufficiently separated angles, and generate multiple optical beams to cover the communication area. With properly designed transmission schemes, it was shown in [19] that the system sum rate increases linearly with the number of UTs. Hence, by using a large transmitter array along with a transmit lens, a BS can simultaneously communicate with a large number of UTs. On the other hand, as indicated above, for a single BS, the channel matrix from the BS to each UT is of rank one, which dramatically limits the transmission rate per UT, so as the throughput of the whole system.

In this paper, we extend single-BS optical MISO wireless communications to networked optical massive MIMO wireless communications, where multiple BSs, each equipped with massive optical transmitters and a transmit lens, simultaneously communicate with a number of UTs, each equipped with multiple photodetectors and a receive lens. We establish the networked optical MIMO channel model, design the transmit covariance matrices, and investigate the sum rate scaling with the number of LEDs. The detailed contributions of this paper are summarized as follows.

- We establish networked optical MIMO channel model. By employing a transmit lens at each BS, lights emitted from different LEDs are refracted to different directions, while with a receive lens at the UT side, lights from different directions are refracted to and received by different photodetectors. Interestingly, for a large number of LEDs, the channel matrix from one BS to different UTs exhibits asymptotically row orthogonality, while for a number of photodetectors, the columns of the channel matrix from different BSs to one UT become orthogonal too. Consequently, each BS is able to transmit independent signals to different UTs and each UT can separate signals from different BSs, which suggests a great potential of networked optical massive MIMO.
- To harvest the full benefits of networked optical massive MIMO, we derive the asymptotic sum rates for a large number of LEDs and/or photodetectors and design the transmit covariance matrices maximizing the sum rate under the total and per LED power constraints. We prove the orthogonality of the optimal transmit covariance matrices, where one LED transmits signals to at most one UT and each UT is served by a non-overlapping

set of LEDs, thus indicating the optimality of beam division multiple access (BDMA) transmission. Furthermore, we propose a simplified BDMA scheme with low complexity, where each BS selects the strongest beam of one UT and transmits the data stream on it, and each UT performs single-user detection with respect to the received signals of the corresponding beams. It is proved that this scheme is asymptotically optimal.

- To gain insight of the asymptotic performance, we analyze the degree of freedom (DoF) in terms of the sum rate scaling with the number of LEDs. Under both power constraints, the DoF of the BDMA transmission is linear in terms of the numbers of BSs and UTs. This indicates that the proposed BDMA transmission can increase the sum rate proportionally to the number of BSs and UTs. Therefore, networked optical massive MIMO communication systems can significantly improve the BS throughput and per UT data rate. Simulation results illustrate that the proposed BDMA transmission can achieve extremely high throughput with low computational complexity and the asymptotically optimal performance.

We adopt the following notations throughout the manuscript: Upper (lower) bold-face letters denote matrices (column vectors); $\mathbf{I}$ denotes the identity matrix, while $\mathbf{1}$ denotes an all-one matrix and $\mathbf{0}$ denotes zero matrix; numeral subscripts of matrices and vectors, if needed, represent their sizes. Also, matrix superscript $(\cdot)^T$ denotes matrix transpose. We use $\text{tr}(\cdot)$ and $\det(\cdot)$ to represent matrix trace and determinant operations, respectively. The inequality $\mathbf{A} \succeq \mathbf{0}$ denotes a positive semi-definite Hermitian matrix $\mathbf{A}$. We use $[\mathbf{A}]_{mn}$ to denote the (m, n) -th element of matrix $\mathbf{A}$.

II. NETWORKED OPTICAL MIMO CHANNEL MODELING

Consider a networked optical massive MIMO system as shown in Fig. 1, where L BSs cooperatively cover a communication area and simultaneously communicate with K UTs. Each BS is equipped with an optical transmitter array with M transmitters, e.g., LEDs or LDs, (we consider LEDs in this paper) and each UT employs a photodetector array with N photodetectors, e.g., avalanche photodiodes (APDs) or PIN photodiodes. In this paper, we focus on optical downlink transmission with LED transmitters at each BS.

Each BS is equipped with a transmit lens such that lights from different LEDs are refracted to different directions, thereby generating multi-directional optical beams. At each UT, a receive lens is used such that lights from different BSs (or different directions) are refracted onto different photodetectors. The receive lens can employ a hemispherical lens [12] or a fisheye lens [11] for high spatial diversity. Fig. 2 illustrates a light propagating from LED j of BS ℓ to photodetector i of UT k . In the following, we will analyze the optical properties of the lens-based transmitter and receiver and establish the channel model of the networked optical massive MIMO system.

A. Optical Analysis of Transmitters

We first consider the refraction of a light from LED j of BS ℓ passing through the transmit lens, as shown in Fig. 2,

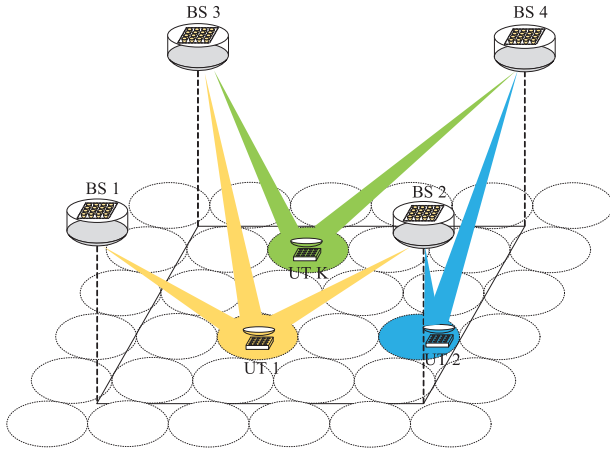
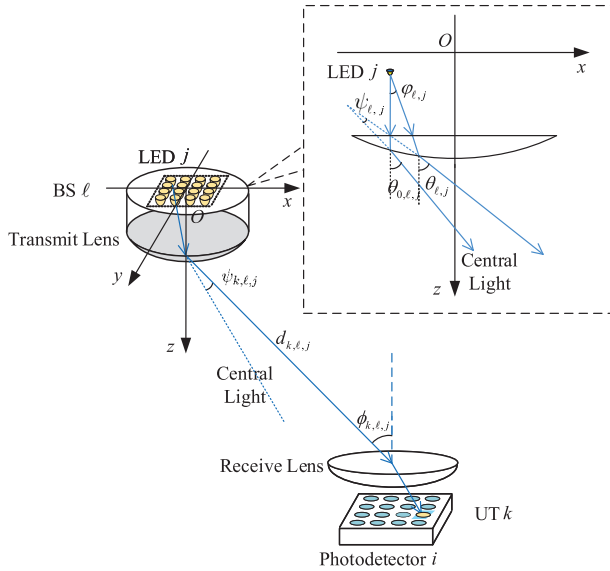


Fig. 1. Networked optical massive MIMO system.

Fig. 2. Schematic diagram of light propagation from BS ℓ to UT k .

where LED j is located at (x_j, y_j, z_j) . A hemispherical lens, consisting of a plane surface and a spherical surface, of refractive index n is installed in front of the LED array. The plane surface is perpendicular to the z axis and located at z_p . The radius of the spherical surface is R , whose center is at the origin of the (x, y, z) coordinate system.

In general, the luminous intensity of an LED follows the Lambertian radiant distribution [10]. Specifically, denote the emission semi-angle of the LEDs as φ_C . Then, the luminous intensity of the light emitted from LED j with polar angle $\varphi_{\ell,j}$ can be expressed as

$$I_0(\varphi_{\ell,j}) = \frac{m_L + 1}{2\pi} \cos^{m_L}(\varphi_{\ell,j}), \quad (1)$$

where $m_L = -\log 2 / \log(\cos(\varphi_C))$ is the order of the Lambertian emission.

The refraction of the emitted light perpendicular to the plane surface of the lens is called the (refracted) central light. The refractive angle between the central light and z axis is given

by

$$\theta_{0,\ell,j} = (n-1) \frac{\sqrt{x_j^2 + y_j^2}}{R}. \quad (2)$$

For the emitted light with polar angle $\varphi_{\ell,j}$, the refractive angle $\theta_{\ell,j}$ between the refracted light and z axis is given in [19], which is rather complex. In practice, we are more concerned about the angle between a refracted light and the central light, i.e., angle $\psi_{\ell,j}$ in Fig. 2, which is referred as the relatively refractive angle. The relation between relatively refractive angle $\psi_{\ell,j}$ and emitted angle $\varphi_{\ell,j}$ can be approximately expressed as [19]

$$\psi_{\ell,j} = \frac{\varphi_{\ell,j}}{n} \left(1 + \frac{nz_j}{R} - \frac{(n-1)^2 z_p}{R} \right). \quad (3)$$

Hence, the ratio of $\psi_{\ell,j}$ to $\varphi_{\ell,j}$ is given by

$$r = \frac{\psi_{\ell,j}}{\varphi_{\ell,j}} = \frac{1}{n} + \frac{z_j(n-1)}{R} - \frac{(n-1)^2 z_p}{nR}. \quad (4)$$

In practice, lights emitted from one LED are generally concentrated within a limited angle. The intensity of a light with a large emitted angle is usually very low or may not impinge on the lens. Thus, we focus on the lights within a limited angle Φ . The optical intensity of the refracted light with relatively refractive angle $\psi_{\ell,j}$ can be expressed by that of the emitting light from LED j with emitted angle $\varphi_{\ell,j}$ in the following form [19]:

$$I_{\ell,j}(\psi_{\ell,j}) = T_{lens} I_0(r^{-1}\psi_{\ell,j}) U(\Phi - r^{-1}\psi_{\ell,j}), \quad (5)$$

where $U(\cdot)$ denotes the unit step function and $T_{lens} \in [0, 1]$ is the optical lens gains.

Remark 1: Optical analysis indicates that the lights emitted from one LED passing through the transmit lens could generate a directional optical beam. Here, we define the beam width ψ_B as the semi-angle at which the optical intensity is half of the maximal intensity. Given that the semi-angle of half intensity of each LED is φ_C , it follows from (4) that the beam width can be expressed as $\psi_B = r\varphi_C$. From (2), the direction of the beam is determined by the horizontal position (x_j, y_j) , while the beam width is determined by the vertical position z_j according to (3). In particular, if $z_j \geq ((n-1)^2 - R)/(n(n-1))$, the ratio r is less than 1 so that the lights from one LED are concentrated within a narrower optical beam after refraction.

B. Optical Analysis of Receivers

Now, we consider lights propagating from BS ℓ to UT k . Particularly, consider a (refracted) light propagating from LED j of BS ℓ to UT k , as shown in Fig. 2. Let $\psi_{k,\ell,j}$ denote the angle between this light and the central light of LED j in BS ℓ , and let $\phi_{k,\ell,j}$ denote the incident angle of this light at UT k . We denote the propagation distance from LED j of BS ℓ to UT k by $d_{k,\ell,j}$ ¹ and denote the total area of the photodetector

¹Here, we consider the distance from the transmit lens at the BS to the receive lens at the UT, while the distance from the LED to the transmit lens is very short and can be neglected. Besides, the effect of the transmit lens has been characterized by the optical intensity $I_{\ell,j}(\psi)$ in (5).

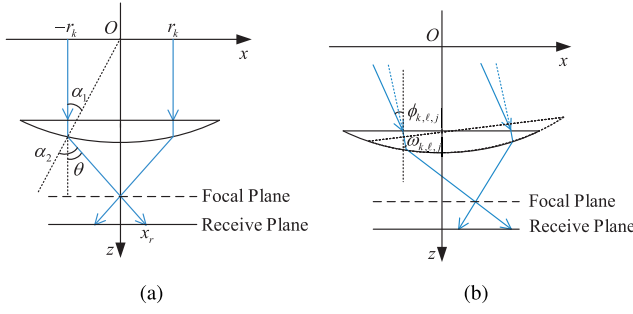


Fig. 3. Schematic diagram of lights passing through the receive lens at the UT side, with the incident angle (a) $\phi_{k,\ell,j} = 0$, and (b) $\phi_{k,\ell,j} \neq 0$.

array at UT k by A_k . Then, the channel gain $g_{k,\ell,j}$ between LED j of BS ℓ and UT k is given by [10]

$$g_{k,\ell,j} = \frac{A_k}{d_{k,\ell,j}^2 r^2} I_{\ell,j}(\psi_{k,\ell,j}) \cos(\phi_{k,\ell,j}). \quad (6)$$

The receive lens at the UT side refracts the incident lights and projects them onto the photodetector array. Note that each photodetector may receive only a part of the optical beam from one LED. The receiving gain of the i th photodetector depends on the ratio between the image area of the i th photodetector and the total image area of the photodetector array. Thus, the channel gain from LED j of BS ℓ to photodetector i of UT k , denoted by $h_{k,i,\ell,j}$, can be presented as [10], [18]

$$h_{k,i,\ell,j} = a_{k,i,\ell,j} g_{k,\ell,j}, \quad (7)$$

where $a_{k,i,\ell,j}$ quantifies how much power falls onto the i th photodetector and is given by

$$a_{k,i,\ell,j} = \frac{A_{k,i,\ell,j}}{A_{k,\ell,j}}, \quad (8)$$

where $A_{k,i,\ell,j}$ denotes the image area of the optical beam of LED j in BS ℓ on photodetector i of UT k , and $A_{k,\ell,j}$ denotes the total image area of the optical beam of LED j on the photodetector array of UT k , where $A_{k,\ell,j} = \sum_i A_{k,i,\ell,j}$.

Remark 2: Note that $g_{k,\ell,j}$ represents the channel gain of the optical beam from LED j of BS ℓ received by the whole photodetector array at UT k , while $h_{k,i,\ell,j}$ represents the channel gain of the i th photodetector. It follows from (7) and (8) that $g_{k,\ell,j} = \sum_i h_{k,i,\ell,j}$. The channel gain $h_{k,i,\ell,j}$ depends on the image area $A_{k,i,\ell,j}$, which, in turn, is related to the light refraction by the receive lens at the UT.

Next, we will analyze the light refraction by the receive lens. As shown in Fig. 3, a receive lens, consisting of a plane surface and a spherical plane, is positioned in front of the receive plane [11], [12]. The radius of the spherical plane is R and the center of the spherical plane is the origin of an orthogonal coordinate system at the UT side. Let the receive aperture of the receive lens be r_k , within which the lights can be refracted and projected onto the receive plane.² The photodetector array is placed on the receive plane and perpendicular to the z axis with position z_r . As the distance between the BS and the UT is much greater than the receive aperture r_k , it is generally

²The receive aperture r_k is no larger than the radius of the plane surface of the receive lens.

assumed that the light rays from an LED arrive at the receive lens with almost the same angle, i.e., the incident lights from one LED are approximately parallel [12], [20], [21].

We first consider the incident light perpendicular to the plane surface of the lens, as shown in Fig. 3(a). As the incident light and the receive lens are rotational symmetry, the image on the receive plane is circular, so we need only to consider the xOz plane. Because the perpendicularly incident light does not refract at the plane surface, we can focus on the light refraction at the spherical plane. Denote the incident angle at the spherical plane by $\alpha_1 = \arcsin(r_k/R)$ and the refractive angle by α_2 . From Snell's law, we have $\sin(\alpha_2) = n \sin(\alpha_1)$. The angle between the refractive light and z axis is given by $\theta = \alpha_2 - \alpha_1$. For small values of α_1 , α_2 and θ , we have $\sin(\theta) \approx \theta$. Substituting $\alpha_2 \approx n\alpha_1$ into the expression of θ , we obtain $\theta \approx (n-1)\alpha_1$. Given the position z_r of the receive plane, the refractive light hits on the receive plane at

$$\begin{aligned} x_r(z_r) &= (z_r - R) \tan(\theta) - R \sin(\alpha_1) \\ &\approx z_r(n-1)\alpha_1 - R\alpha_1. \end{aligned} \quad (9)$$

Hence, the image on the receive plane is a circle with radius $|x_r(z_r)|$, which can be expressed as $\mathcal{I} = \{(x, y) | x^2 + y^2 \leq x_r(z_r)^2\}$.

When the receive plane is located at $z_r = \frac{nR}{n-1}$, the refractive light will be focused at the point $x_r = 0$, which is independent of the incident angle α_1 . Such a point is called the focus point. In this case, the distance between the receive lens and the receive plane $z_r - R = R/(n-1) = f$ is called the focal length.

When the incident light is oblique with angle $\phi_{k,\ell,j}$, as shown in Fig. 3(b), the refractive angle at the plane surface is $\omega_{k,\ell,j}$, satisfying the relationship $\sin(\phi_{k,\ell,j}) = n \sin(\omega_{k,\ell,j})$. By virtually rotating the receive lens $\omega_{k,\ell,j}$ degrees with respect to the origin point, the light passing through the plane surface can be treated as the incident light perpendicular to the rotated virtual lens. Then, we are able to calculate the image area analytically. Specifically, when $z_r \leq nR/(n-1)$, the image area can be expressed as

$$\mathcal{I}_{k,\ell,j} = \left\{ (x, y) \left| \begin{aligned} -r_k + (z_r - R) \tan(\theta + \omega_{k,\ell,j}) &\leq x, \\ x &\leq r_k - (z_r - R) \tan(\theta - \omega_{k,\ell,j}), \\ -d(x, \omega_{k,\ell,j}) &\leq y \leq d(x, \omega_{k,\ell,j}) \end{aligned} \right. \right\}, \quad (10)$$

and when $z_r > nR/(n-1)$, the image area is given by

$$\mathcal{I}_{k,\ell,j} = \left\{ (x, y) \left| \begin{aligned} r_k - (z_r - R) \tan(\theta - \omega_{k,\ell,j}) &\leq x, \\ x &\leq -r_k + (z_r - R) \tan(\theta + \omega_{k,\ell,j}), \\ -d(x, \omega_{k,\ell,j}) &\leq y \leq d(x, \omega_{k,\ell,j}) \end{aligned} \right. \right\}, \quad (11)$$

where $d(x, \omega_{k,\ell,j}) = \sqrt{(x_r(d_1(x, \omega_{k,\ell,j}))^2 - d_2(x, \omega_{k,\ell,j}))^2}$, and $d_1(x, \omega_{k,\ell,j})$ and $d_2(x, \omega_{k,\ell,j})$ are given by

$$\begin{aligned} d_1(x, \omega_{k,\ell,j}) &= (z_r + x \tan(\omega_{k,\ell,j})) \cos(\omega_{k,\ell,j}) \\ d_2(x, \omega_{k,\ell,j}) &= z_r \sin(\omega_{k,\ell,j}) - x \cos(\omega_{k,\ell,j}). \end{aligned} \quad (12)$$

Suppose the photodetector array is a square at the receive plane with $N = S \times S$ photodetectors. Let the side length of

the array be D . The area of the i th photodetector is

$$\mathcal{A}_i = \left\{ (x, y) \left| \begin{array}{l} -D/2 + (i_1 - 1)D/S \leq x, \\ x \leq -D/2 + i_1 D/S, \\ -D/2 + (i_2 - 1)D/S \leq y, \\ y \leq -D/2 + i_2 D/S \end{array} \right. \right\}, \quad (13)$$

where $i = (i_1 - 1)S + i_2$ and $1 \leq i_1, i_2 \leq S$ are the row and column indices in the photodetector array. Then, the total image area $A_{k,\ell,j}$ and the image area on the i th photodetector $A_{k,i,\ell,j}$ can be respectively calculated as

$$A_{k,\ell,j} = \int_{A \in \mathcal{I}_{k,\ell,j}} dA, \quad A_{k,i,\ell,j} = \int_{A \in \mathcal{I}_{k,\ell,j} \cap \mathcal{A}_i} dA. \quad (14)$$

Thus, from (8), the coefficient $a_{k,i,\ell,j}$ is given by

$$a_{k,i,\ell,j} = \frac{\int_{A \in \mathcal{I}_{k,\ell,j} \cap \mathcal{A}_i} dA}{\int_{A \in \mathcal{I}_{k,\ell,j}} dA}. \quad (15)$$

Remark 3: The coefficient $a_{k,i,\ell,j} \in [0, 1]$ represents the ratio of the received optical power on photodetector i to the total received optical power. If the refracted lights are concentrated within one photodetector, i.e., $\mathcal{I}_{k,\ell,j} \subseteq \mathcal{A}_i$, then $a_{k,i,\ell,j} = 1$. From (10) and (11), the image area depends on the incident angle $\phi_{k,\ell,j}$. If the LEDs are sufficiently separated, the images of the optical beams from different LEDs will separate and fall onto different photodetectors [11].

C. Channel Properties

The LED array in each BS generally consists of a number of compactly packaged LED chips. In practice, the size of an LED array is much smaller than the propagation distance from a BS to a UT. As a consequence, the incident angles of the optical beams from different LEDs in an LED array at a UT are approximately identical [10], [12], [21]. Specifically, consider the incident angle $\phi_{k,\ell,j}$ of the optical beam from LED j in BS ℓ to UT k in Fig. 3, which is then invariant with respect to j . Thus, from (10) and (11), the images $\mathcal{I}_{k,\ell,j}$ are completely overlapping, and the coefficients $a_{k,i,\ell,j}$ are independent of j , i.e., $a_{k,i,\ell,j} = a_{k,i,\ell}, \forall j$. Let $\mathbf{a}_{k,\ell} = [a_{k,1,\ell}, a_{k,2,\ell}, \dots, a_{k,N,\ell}]^T \in \mathbb{R}^{N \times 1}$ be the receive coefficient vector of the optical beams from BS ℓ , and $\mathbf{g}_{k,\ell} = [g_{k,\ell,1}, g_{k,\ell,2}, \dots, g_{k,\ell,M}]^T \in \mathbb{R}^{M \times 1}$ be the vector of the channel gains from different LEDs in BS ℓ to the receive lens at UT k . Then, according to (7), the channel matrix from BS ℓ to UT k can be expressed as

$$\mathbf{H}_{k,\ell} = \mathbf{a}_{k,\ell} \mathbf{g}_{k,\ell}^T. \quad (16)$$

Remark 4: The above model implies that, though the BS and the UT are equipped with LED and photodetector arrays respectively, the channel matrix between one BS and one UT is of rank one. This is because, as pointed above, the received optical beams from the LEDs in one array are approximately parallel, as a result of the compact size of the LED array and the relatively large distance between the BS and UT. Such a rank-one channel property has been commonly observed in optical wireless communications [10], [12]. The rank deficiency implies that one BS can only transmit one stream to

each UT, thus limiting data rates. To support multi-stream transmission [22], we consider multiple separated BSs and propose networked optical massive MIMO to generate higher degrees of spatial freedom.

Consider the networked optical massive MIMO system as shown in Fig. 1. The channel matrix from L BSs to UT k can be expressed as

$$\mathbf{H}_k = [\mathbf{a}_{k,1} \mathbf{g}_{k,1}^T, \mathbf{a}_{k,2} \mathbf{g}_{k,2}^T, \dots, \mathbf{a}_{k,L} \mathbf{g}_{k,L}^T], \quad (17)$$

where the ℓ th sub-matrix represents the channel gains from BS ℓ to UT k . In the following, we show that, as the number of LEDs or/and photodetectors increases, the channel matrix $\mathbf{H}_k$ exhibits some favorable asymptotic properties.

At the BS side, when the number of LEDs grows to infinity, the LED array should be located at the focal plane of the transmit lens and the lights emitted from one LED are refracted to one direction [19]. Thus, the optical intensity of the refracted light with relatively refractive angle $\psi_{k,\ell,j}$ reduces to

$$\lim_{M \rightarrow \infty} I_{\ell,j}(\psi_{k,\ell,j}) = T_{lens} \frac{m_L + 1}{2\pi} \delta(\psi_{k,\ell,j}). \quad (18)$$

Therefore, light intensities from one LED to different UTs tend to be orthogonal, resulting in the row orthogonality of the channel matrices from one BS to different UTs, i.e.,

$$\lim_{M \rightarrow \infty} \mathbf{H}_{k_1,\ell} \mathbf{H}_{k_2,\ell}^T = \mathbf{0}, \quad k_1 \neq k_2. \quad (19)$$

This result reveals that the row vectors of channel matrices from BS ℓ to different UTs become asymptotically orthogonal, implying that the signals intended to one UT are not received by other UTs. Then, the channel matrices from L BSs to different UTs become row orthogonal, i.e., $\lim_{M \rightarrow \infty} \mathbf{H}_{k_1} \mathbf{H}_{k_2}^T = \mathbf{0}$. Therefore, BSs have the potential to simultaneously transmit independent signals to different UTs without interference, which is particularly desirable for multi-user communications.

At the UT side, when the photodetector array is located at the focal plane of the receive lens, the (parallel) lights from one BS are focused within a small spot on the photodetector array. Hence, the image areas of the optical beams from the sufficiently separated BSs are non-overlapping. Furthermore, when the photodetector array in the UT contains a large number of photodetectors, each photodetector is of a shrunk size and apt to receive lights from at most one BS, and then lights from different BSs fall onto different photodetectors. Hence, as the number of photodetectors increases to infinity, the receive vectors $\mathbf{a}_{k,\ell_1}$ and $\mathbf{a}_{k,\ell_2}$ become orthogonal, i.e., $\lim_{N \rightarrow \infty} \mathbf{a}_{k,\ell_1}^T \mathbf{a}_{k,\ell_2} = 0$, resulting in the column orthogonality of channel matrices from different BSs to one UT, given by

$$\lim_{N \rightarrow \infty} \mathbf{H}_{k,\ell_1}^T \mathbf{H}_{k,\ell_2} = \mathbf{0}, \quad \ell_1 \neq \ell_2. \quad (20)$$

In practice, with limited photodetectors and a well designed receive lens, the channel matrices $\mathbf{H}_{k,\ell_1}$ and $\mathbf{H}_{k,\ell_2}$ can also be approximately orthogonal [11], [23].

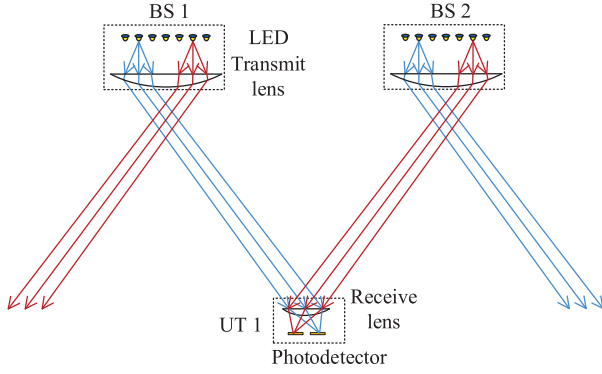


Fig. 4. The schematic diagram of the asymptotic properties.

This result indicates that with a number of photodetectors, different photodetectors receive signals from different BSs and one UT can separate signals from different BSs without interference. In this case, the matrix $\mathbf{H}_k^T \mathbf{H}_k$ becomes a block diagonal matrix, given by

$$\lim_{N \rightarrow \infty} \mathbf{H}_k^T \mathbf{H}_k = \text{bdiag}\{\mathbf{g}_{k,1} \mathbf{a}_{k,1}^T \mathbf{a}_{k,1} \mathbf{g}_{k,1}^T, \mathbf{g}_{k,2} \mathbf{a}_{k,2}^T \mathbf{a}_{k,2} \mathbf{g}_{k,2}^T, \dots, \mathbf{g}_{k,L} \mathbf{a}_{k,L}^T \mathbf{a}_{k,L} \mathbf{g}_{k,L}^T\}. \quad (21)$$

Suppose that UTs are cooperatively served by L BSs and can receive signals from them, i.e., $\mathbf{a}_{k,\ell} \mathbf{g}_{k,\ell}^T \neq \mathbf{0}, \forall \ell$. Then, the rank of channel matrix $\mathbf{H}_k$ tends to

$$\lim_{N \rightarrow \infty} \text{rank}(\mathbf{H}_k) = \lim_{N \rightarrow \infty} \text{rank}(\mathbf{H}_k^T \mathbf{H}_k) = L, \quad (22)$$

which reveals that each UT can receive and separate L independent data streams.

Remark 5: As the number of LEDs increases, the channel matrices from one BS to different UTs become row orthogonal, while with a large number of photodetectors, the channel matrices from different BSs to one UT are column orthogonal. In practice, as BSs are separated sufficiently apart, the asymptotic properties in (21) and (22) can be achieved with fewer number of photodetectors [11], [12]. The small-scale LEDs and photodetectors can still mitigate the interference, as shown in the simulation results, owing to the approximately orthogonal properties of the channel matrices. The channel orthogonality indicates that different LEDs at each BS generate beams towards different directions, while the received signals at the UT from different directions are received by different photodetectors, as shown in Fig. 4. The channel orthogonality plays an important role in RF massive MIMO communications [17], whereas this favorable property has not been found in optical wireless communications until recently. In this paper, we discover the asymptotic channel orthogonality in networked optical massive MIMO communications, and show that using a large number of LEDs and photodetectors facilitates reducing the interference between UTs and the interference between BSs, respectively. Therefore, networked optical massive MIMO communications have the potential to significantly improve the BS throughput and per UT transmission rate.

III. NETWORKED BDMA TRANSMISSION UNDER TOTAL POWER CONSTRAINT

In this section, we develop proper networked optical massive MIMO transmission schemes based on the above established channel model. Specifically, consider an optical massive MIMO communication system, where L BSs simultaneously transmit signals to K UTs. The BS encodes the information of each UT independently for transmission. Specifically, denote the transmitted signal intended to UT k as $\mathbf{x}_k = [\mathbf{x}_{k,1}^T, \mathbf{x}_{k,2}^T, \dots, \mathbf{x}_{k,L}^T]^T$, where the transmitted signal from BS ℓ , denoted by $\mathbf{x}_{k,\ell} \in \mathbb{R}^{M \times 1}$, is zero-mean with covariance $\mathbf{Q}_{k,\ell} = \mathbb{E}\{\mathbf{x}_{k,\ell} \mathbf{x}_{k,\ell}^T\}$. Then, the covariance matrix of $\mathbf{x}_k$ can be expressed as $\mathbf{Q}_k = \mathbb{E}\{\mathbf{x}_k \mathbf{x}_k^T\} = \text{bdiag}\{\mathbf{Q}_{k,1}, \mathbf{Q}_{k,2}, \dots, \mathbf{Q}_{k,L}\}$. A direct current (DC) bias is added to guarantee a non-negative input signal to the LEDs. After removing the DC bias, the received signal at UT k can be written as

$$\mathbf{y}_k = \mathbf{H}_k \mathbf{x}_k + \sum_{k' \neq k} \mathbf{H}_k \mathbf{x}_{k'} + \mathbf{n}_k = \mathbf{H}_k \mathbf{x}_k + \mathbf{z}_k, \quad (23)$$

where $\mathbf{n}_k$ is the noise vector, whose elements are independently and identically distributed (i.i.d.) Gaussian variables with zero mean and variance σ^2 , $\mathbf{z}_k = \sum_{k' \neq k} \mathbf{H}_k \mathbf{x}_{k'} + \mathbf{n}_k$ is the aggregated interference plus noise, which is usually treated as the Gaussian noise with covariance matrix $\mathbf{K}_k = \sum_{k' \neq k} \mathbf{H}_k \mathbf{Q}_{k'} \mathbf{H}_k^T + \sigma^2 \mathbf{I}$ [24]. Without loss of generality, we assume a unit noise variance (i.e., $\sigma^2 = 1$).

A. Asymptotic Sum Rate

In general, as the transmitted signals in optical wireless communications are limited, there is no closed-form expression for the capacity. Fortunately, a lower bound and an upper bound of the transmission rate R_k can be found [25], [26]

$$R_k \geq \frac{1}{2} \log \det \left(\mathbf{I} + \frac{6}{e\pi} \mathbf{K}_k^{-1} \mathbf{H}_k \mathbf{Q}_k \mathbf{H}_k^T \right) \quad (24)$$

$$R_k \leq \frac{1}{2} \log \det \left(\mathbf{I} + \mathbf{K}_k^{-1} \mathbf{H}_k \mathbf{Q}_k \mathbf{H}_k^T \right), \quad (25)$$

where the lower bound is derived from the uniform distribution, and the upper bound is derived from the Gaussian distribution. Note that the difference between the expressions of the lower and upper bounds is the coefficient constant $6/e\pi$ in (24) in front of $\mathbf{K}_k^{-1} \mathbf{H}_k \mathbf{Q}_k \mathbf{H}_k^T$. Hence, we introduce a coefficient γ and define the following approximate transmission rate

$$R_k = \frac{1}{2} \log \det \left(\mathbf{I} + \gamma \mathbf{K}_k^{-1} \mathbf{H}_k \mathbf{Q}_k \mathbf{H}_k^T \right). \quad (26)$$

which contains the lower and upper bounds as the special cases, and coincides with a fitting function of the capacity in [27] as well. Hence, R_k can be regarded as a general expression of the achievable rate of the system.

Recall from (17) that the channel matrix $\mathbf{H}_{k,\ell}$ can be decomposed as $\mathbf{H}_{k,\ell} = \mathbf{a}_{k,\ell} \mathbf{g}_{k,\ell}^T$. Let $\mathbf{C}_{k,\ell} = \mathbf{a}_{k,\ell} \mathbf{a}_{k,\ell}^T$, and

then the sum rate can be written as

$$\begin{aligned}
 R_{\text{sum}} &= \frac{1}{2} \sum_k \log \det \left(\mathbf{I} + \sum_{\ell} \mathbf{g}_{k,\ell}^T \left(\gamma \mathbf{Q}_{k,\ell} + \sum_{k' \neq k} \mathbf{Q}_{k',\ell} \right) \right. \\
 &\quad \left. \times \mathbf{g}_{k,\ell} \mathbf{C}_{k,\ell} \right) \\
 &\quad - \frac{1}{2} \sum_k \log \det \left(\mathbf{I} + \sum_{\ell} \mathbf{g}_{k,\ell}^T \left(\sum_{k' \neq k} \mathbf{Q}_{k',\ell} \right) \mathbf{g}_{k,\ell} \mathbf{C}_{k,\ell} \right).
 \end{aligned} \tag{27}$$

Employing the similar method in [19], we can derive the asymptotic value of the sum rate $\bar{R}_{\text{sum}}$.

Theorem 1: As M grows asymptotically to infinity, the achievable sum rate R_{sum} tends to $\bar{R}_{\text{sum}}$, i.e.,

$$\lim_{M \rightarrow \infty} (R_{\text{sum}} - \bar{R}_{\text{sum}}) = 0, \tag{28}$$

where $\bar{R}_{\text{sum}}$ is given by

$$\begin{aligned}
 \bar{R}_{\text{sum}} &= \frac{1}{2} \sum_k \log \det \left(\mathbf{I} + \sum_{\ell} M^2 c_{k,\ell}^2 \left(\gamma [\mathbf{Q}_{k,\ell}]_{m_{k,\ell} m_{k,\ell}} \right. \right. \\
 &\quad \left. \left. + \sum_{k' \neq k} [\mathbf{Q}_{k',\ell}]_{m_{k,\ell} m_{k,\ell}} \right) \mathbf{C}_{k,\ell} \right) \\
 &\quad - \frac{1}{2} \sum_k \log \det \left(\mathbf{I} + \sum_{\ell} M^2 c_{k,\ell}^2 \left(\sum_{k' \neq k} [\mathbf{Q}_{k',\ell}]_{m_{k,\ell} m_{k,\ell}} \right) \right. \\
 &\quad \left. \times \mathbf{C}_{k,\ell} \right),
 \end{aligned} \tag{29}$$

where $m_{k,\ell}$ satisfies $\psi_{k,\ell,m_{k,\ell}} = 0$ and

$$c_{k,\ell} = T_{\text{lens}} \frac{A_k}{r^2 d_{k,\ell}^2} \frac{(m_L + 1)}{2\pi} \cos(\phi_{k,\ell}). \tag{30}$$

Remark 6: Theorem 1 presents an asymptotic sum rate $\bar{R}_{\text{sum}}$ with an infinite number of LEDs. With limited LEDs, $\bar{R}_{\text{sum}}$ is an approximation, where $m_{k,\ell}$ is selected according to the smallest angle $|\psi_{k,\ell,m_{k,\ell}}|$. This indicates that each UT receives signals from one dominant LED in an LED array. Interestingly, from (29), the asymptotic sum rate $\bar{R}_{\text{sum}}$ only depends on the diagonal elements of $\mathbf{Q}_{k,\ell}$, implying that for a large number of LEDs, the transmit covariance matrix design can be simplified into designing a number of diagonal elements.

B. Optimal Design With Asymptotically Large Number of LEDs

Next, we will design the transmit covariance matrices $\mathbf{Q}_{k,\ell}$ to maximize the asymptotic sum rate $\bar{R}_{\text{sum}}$. In this section, we consider the typical total power constraint, $\sum_k \text{tr}(\mathbf{Q}_{k,\ell}) \leq P_{\ell}$, where P_{ℓ} is the total power of BS ℓ . The transmission design problem under the total power constraint

can be formulated as

$$\begin{aligned}
 \max_{\mathbf{Q}_{k,\ell}} \quad & \bar{R}_{\text{sum}} \\
 \text{s.t.} \quad & \sum_k \text{tr}(\mathbf{Q}_{k,\ell}) \leq P_{\ell}, \quad \mathbf{Q}_{k,\ell} \succeq \mathbf{0}.
 \end{aligned} \tag{31}$$

Before solving problem (31) directly, we provide the following result to simplify it.

Theorem 2: The optimal covariance matrix $\mathbf{Q}_{k,\ell}$ is a diagonal matrix, and the diagonal elements satisfy

$$\mathbf{Q}_{k_1,\ell} \mathbf{Q}_{k_2,\ell} = \mathbf{0}, \quad k_1 \neq k_2. \tag{32}$$

Proof: See Appendix A. ■

Remark 7: The diagonal covariance matrix reveals that different LEDs transmit independent signals. Furthermore, the transmit covariance matrices from one BS to different UTs should be orthogonal. This result indicates that one LED transmits signals to at most one UT and each UT is served by a non-overlapping set of LEDs. Such a transmission scheme is referred to as BDMA [28], while Theorem 2 indicates that BDMA is asymptotically optimal for the networked optical massive MIMO communication system.

Under condition (32), the asymptotic sum rate can be simplified as

$$\bar{R}_{\text{sum}} = \frac{1}{2} \sum_k \log \det \left(\mathbf{I} + \gamma \sum_{\ell} M^2 c_{k,\ell}^2 [\mathbf{Q}_{k,\ell}]_{m_{k,\ell} m_{k,\ell}} \mathbf{C}_{k,\ell} \right). \tag{33}$$

Let $q_{k,\ell} = [\mathbf{Q}_{k,\ell}]_{m_{k,\ell} m_{k,\ell}}$ represent the transmit power from the $m_{k,\ell}$ th beam of BS ℓ to UT k and $\mathbf{h}_{k,\ell} = \sqrt{\gamma} M c_{k,\ell} \mathbf{a}_{k,\ell}$ represent the channel vector from the $m_{k,\ell}$ th beam to UT k . Then, the sum rate (33) can be rewritten as

$$\bar{R}_{\text{sum}} = \frac{1}{2} \sum_k \log \det \left(\mathbf{I} + \sum_{\ell} q_{k,\ell} \mathbf{h}_{k,\ell} \mathbf{h}_{k,\ell}^T \right), \tag{34}$$

and the transmit covariance matrix design problem in (31) reduces to the following power allocation problem:

$$\begin{aligned}
 \max_{q_{k,\ell}} \quad & \frac{1}{2} \sum_k \log \det \left(\mathbf{I} + \sum_{\ell} q_{k,\ell} \mathbf{h}_{k,\ell} \mathbf{h}_{k,\ell}^T \right) \\
 \text{s.t.} \quad & \sum_k q_{k,\ell} \leq P_{\ell}, \quad q_{k,\ell} \geq 0.
 \end{aligned} \tag{35}$$

The following result provides an analytical solution to problem (35).

Theorem 3: The optimal power allocation $q_{k,\ell}$ is given by

$$q_{k,\ell} = \left(\frac{1}{\eta_{\ell}} - \frac{1}{\mathbf{h}_{k,\ell}^T \left(\mathbf{I} + \sum_{j \neq \ell} q_{k,j} \mathbf{h}_{k,j} \mathbf{h}_{k,j}^T \right)^{-1} \mathbf{h}_{k,\ell}} \right)^+, \tag{36}$$

where $(x)^+ = \max\{x, 0\}$, and η_{ℓ} satisfies the power constraint $\sum_k q_{k,\ell} = P_{\ell}$.

Proof: See Appendix B. ■

The optimal power allocation maximizing the asymptotic sum rate has a similar structure with the water-filling solution. The difference is that due to the summation in sum rate (34),

the transmit power $q_{k,\ell}$ of BS ℓ is coupled with the transmit power $q_{k,j}$ of BS j , which makes it not straightforward to calculate the optimal $q_{k,\ell}$. In the following, we propose a water-filling based power allocation algorithm that iteratively compute each $q_{k,\ell}$ while fixing the transmit powers of the other BSs.

Algorithm 1 Water-Filling Based Power Allocation Algorithm

- 1: Initialize $q_{k,j}^{(0)} = 0, i = 0$.
- 2: **repeat**
- 3: **for** $\ell = 1$ to L **do**
- 4: Employ the water-filling algorithm to calculate $q_{k,\ell}^{(i)}$ according to

$$q_{k,\ell}^{(i+1)} = \left((\eta_\ell^{(i+1)})^{-1} - \left(\mathbf{h}_{k,\ell}^T \left(\mathbf{I} + \sum_{j=1}^{\ell-1} q_{k,j}^{(i+1)} \mathbf{h}_{k,j} \mathbf{h}_{k,j}^T \right)^{-1} \mathbf{h}_{k,\ell} \right)^{-1} \right)^+ + \sum_{j=\ell+1}^L q_{k,j}^{(i)} \mathbf{h}_{k,j} \mathbf{h}_{k,j}^T \right)^{-1} \mathbf{h}_{k,\ell} \right)^{-1} \right)^+,$$

where $\eta_\ell^{(i+1)}$ is chosen satisfying the power constraint.

- 5: **end for**
 - 6: Set $i = i + 1$ and calculate $\bar{R}_{\text{sum}}^{(i)}$.
 - 7: **until** $|\bar{R}_{\text{sum}}^{(i)} - \bar{R}_{\text{sum}}^{(i-1)}| \leq \epsilon$.
-

Remark 8: In each iteration, Algorithm 1 sequentially calculates the power allocation of each BS. Specifically, the power allocation of the ℓ th BS in the $(i+1)$ th iteration is updated according to those of the $(\ell-1)$ BSs that have been updated in the $(i+1)$ th iteration as well as those of the left BSs that keep the values in the i th iteration. Under this approach, in the each iteration, the sum rate will monotonically increase. Hence, the sum rate sequence $\{\bar{R}_{\text{sum}}^{(i)}\}$ generated by Algorithm 1 converges. Since the water filling algorithm needs $O(K)$ arithmetical and logical operations [29], the complexity of Algorithm 1 is $O(LK)$.

C. Optimal Design With Asymptotically Large Number of LEDs and Photodetectors

As the number of optical receivers tends to infinity, the receive vectors $\mathbf{a}_{k,\ell_1}$ and $\mathbf{a}_{k,\ell_2}$ become orthogonal (i.e., $\lim_{N \rightarrow \infty} \mathbf{a}_{k,\ell_1}^T \mathbf{a}_{k,\ell_2} = 0$), and we have

$$\lim_{N \rightarrow \infty} \mathbf{C}_{k,\ell_1} \mathbf{C}_{k,\ell_2} = \mathbf{0}. \quad (37)$$

Then, the asymptotic sum rate (29) can be further simplified as

$$\bar{R}_{\text{sum}} = \frac{1}{2} \sum_k \sum_\ell \log(1 + \gamma M^2 a_{k,\ell} c_{k,\ell}^2 [\mathbf{Q}_{k,\ell}]_{m_{k,\ell} m_{k,\ell}}), \quad (38)$$

where $a_{k,\ell} = \mathbf{a}_{k,\ell}^T \mathbf{a}_{k,\ell}$, and problem (31) can be degraded to

$$\begin{aligned} \max_{\mathbf{Q}_{k,\ell}} & \quad \frac{1}{2} \sum_k \sum_\ell \log(1 + \gamma M^2 a_{k,\ell} c_{k,\ell}^2 [\mathbf{Q}_{k,\ell}]_{m_{k,\ell} m_{k,\ell}}) \\ \text{s.t.} & \quad \sum_k \text{tr}(\mathbf{Q}_{k,\ell}) \leq P_\ell, \quad \mathbf{Q}_{k,\ell} \succeq \mathbf{0}. \end{aligned} \quad (39)$$

For problem (39), the optimal $\mathbf{Q}_{k,\ell}$ can be obtained as follows.

Theorem 4: The optimal covariance matrix $\mathbf{Q}_{k,\ell}$ is

$$[\mathbf{Q}_{k,\ell}]_{m_{k,\ell} m_{k,\ell}} = \begin{cases} \left(\frac{1}{\mu_\ell} - \frac{1}{\gamma M^2 a_{k,\ell} c_{k,\ell}^2} \right)^+, & k = \arg \max_{k'} a_{k',\ell} c_{k',\ell}^2, \\ 0, & k \neq \arg \max_{k'} a_{k',\ell} c_{k',\ell}^2, \end{cases} \quad (40)$$

where $(x)^+ = \max\{x, 0\}$, μ_ℓ is the Lagrange multiplier satisfying the condition

$$\sum_k [\mathbf{Q}_{k,\ell}]_{m_{k,\ell} m_{k,\ell}} = P_\ell. \quad (41)$$

Remark 9: Solution (40) is a special case of result (36). With a large number of photodetectors at the UT side, the receive vectors $\mathbf{a}_{k,\ell_1}$ and $\mathbf{a}_{k,\ell_2}$ become orthogonal and each UT can separate signals from different BSs. From (38), with a large number of LEDs and photodetectors, the sum rate is the summation of KL single-user single-input single-output (SU-SISO) rates without interference. In this case, the optimal transmit strategy is that each BS selects the strongest beam of one UT and transmit one data stream on it, and the optimal receive strategy is that each UT performs single-user detection with respect to the received signal of one BS. Thus, the transceiver design is simple and easy to accomplish.

D. Sum Rate Scaling Analysis

To further reveal the insight of the asymptotic performance of the networked optical massive MIMO system, we investigate the sum rate scaling with the number of LEDs by analyzing the DoF. In conventional communication systems, DoF is typically defined as the limit of the ratio of sum rate $\bar{R}_{\text{sum}}$ to $\log(SNR)$ [30], i.e., $\lim_{SNR \rightarrow \infty} \bar{R}_{\text{sum}} / \log(SNR)$, where SNR is the ratio of the transmitted signal's power to the noise power. In massive MIMO communication systems, it is more meaningful to specify how the sum rate scales with the log of the number of transmit antennas, which leads to another definition of DoF [31]

$$\text{DoF} = \lim_{M \rightarrow \infty} \frac{\bar{R}_{\text{sum}}}{\log(M)}. \quad (42)$$

Here, we use the DoF definition in (42) for optical massive MIMO wireless communications.

For comparison, we first consider the single-user transmission case, where a BS (BS 1) transmits signal to a UT (UT 1) without lens at both the BS and UT sides. Then, the channel matrix $\tilde{\mathbf{H}}$ reduces to $\tilde{\mathbf{H}} = h \mathbf{1}_{N \times M}$, where $h = \frac{A_1/N}{d_{1,1}^2} I_0(\phi_{1,1}) \cos(\phi_{1,1})$ represents the channel gain coefficient from BS 1 to UT 1. The achievable rate of UT 1 is given by

$$R_1 = \frac{1}{2} \log \det(\mathbf{I} + \gamma \tilde{\mathbf{H}} \mathbf{Q} \tilde{\mathbf{H}}^T). \quad (43)$$

To maximize the single user rate R_1 under the total power constraint, the optimal transmit covariance matrix should be $\mathbf{Q} = P/M \mathbf{1}_{M \times M}$, and the maximal single user rate is

$$R_1^o = \frac{1}{2} \log(1 + \gamma h^2 M N P). \quad (44)$$

Thus, the degree of freedom is

$$DoF_1 = \lim_{M \rightarrow \infty} \frac{R_1^o}{\log(M)} = \frac{1}{2}. \quad (45)$$

According to (38) and the definition of DoF (42), the DoF of BDMA transmission is presented as follows.

Theorem 5: The DoF of BDMA transmission under the total power constraint is

$$\begin{aligned} DoF &= \lim_{M \rightarrow \infty} \frac{\frac{1}{2} \sum_k \sum_\ell \log \left(1 + \gamma M^2 a_{k,\ell} c_{k,\ell}^2 [\mathbf{Q}_{k,\ell}]_{m_{k,\ell} m_{k,\ell}} \right)}{\log(M)} \\ &= KL. \end{aligned} \quad (46)$$

Remark 10: The DoF represents the multiplexing gain of the optical communication system with a large number of LEDs. Theorem 5 indicates that the proposed BDMA transmission can increase the sum rate proportionally to the number of BSs and UTs. Moreover, compared with the single-user rate in (44), we have

$$\begin{aligned} &\lim_{M \rightarrow \infty} \frac{\bar{R}_{\text{sum}}}{R_1^o} \\ &= \lim_{M \rightarrow \infty} \frac{\frac{1}{2} \sum_k \sum_\ell \log \left(1 + \gamma M^2 a_{k,\ell} c_{k,\ell}^2 [\mathbf{Q}_{k,\ell}]_{m_{k,\ell} m_{k,\ell}} \right)}{\frac{1}{2} \log(1 + \gamma h^2 MNP)} \\ &= 2KL, \end{aligned} \quad (47)$$

where the coefficient 2 is the power gain due to the lens and KL represents the multiplexing gain of the networked BDMA transmission. Specifically, each BS has the ability to communicate with L UTs simultaneously, while each UT can receive and separate the signals from K BSs. Therefore, networked optical BDMA transmission significantly improves the BS throughput as well as the transmission rate per UT.

IV. NETWORKED BDMA TRANSMISSION UNDER PER LED POWER CONSTRAINT

In optical communications, the common IM/DD schemes require driving current must be non-negative. Hence, the transmitted current of each LED is limited to guarantee the non-negative input signal. With bounded data signals, this amplitude constraint can be realized via the per LED power constraint [15], [19], where the transmit power of each LED is constrained to limit the amplitude of the signal.

A. Asymptotic Optimal Design

Mathematically, the per LED power constraint can be formulated as $\sum_k [\mathbf{Q}_{k,\ell}]_{mm} \leq p$, where p is the maximal power of each LED. Then, the transmit covariance matrix design problem under the per LED power constraint can be expressed as

$$\begin{aligned} &\max_{\mathbf{Q}_{k,\ell}} \bar{R}_{\text{sum}} \\ &\text{s.t.} \quad \sum_k \mathbf{e}_m^T \mathbf{Q}_{k,\ell} \mathbf{e}_m \leq p, \quad \mathbf{Q}_{k,\ell} \succeq \mathbf{0}, \end{aligned} \quad (48)$$

where $\mathbf{e}_m$ is a unit vector with the m th element being 1 and others being zeros. The solution to problem (48) is provided in the following result.

Theorem 6: The optimal covariance matrix $\mathbf{Q}_{k,\ell}$ is a diagonal matrix, and the diagonal elements are given by

$$[\mathbf{Q}_{k,\ell}]_{mm} = \begin{cases} p, & m = m_{k,\ell} \\ 0, & m \neq m_{k,\ell}, \end{cases} \quad (49)$$

where $m_{k,\ell}$ satisfies $\psi_{k,\ell,m_{k,\ell}} = 0$.

Proof: See Appendix C. ■

Remark 11: Under the per LED power constraint, the optimal transmission strategy is that different LEDs transmit independent signals, and the transmitted beams for different UTs are non-overlapping, similar as in the case of the total power constraint. On the other hand, under the per LED power constraint, the optimal power allocation is that each LED uses the maximal power p to transmit the signals.

When BSs transmit signals with the optimal covariance matrix $\mathbf{Q}_{k,\ell}$ in Theorem 6, the asymptotic sum rate can be expressed as

$$\bar{R}_{\text{sum}} = \frac{1}{2} \sum_k \log \det \left(\mathbf{I} + \gamma \sum_\ell M^2 c_{k,\ell}^2 p \mathbf{C}_{k,\ell} \right). \quad (50)$$

From (50), when BSs are equipped with a large number of LEDs, the sum rate is equal to the summation of K single-users' rate. To achieve sum rate (50), each BS employs one beam to transmit an independent signal to one UT, which significantly reduces the complexity of the transmit design.

Moreover, when the number of photodetectors is large enough, the receive vectors $\mathbf{a}_{k,\ell_1}$ and $\mathbf{a}_{k,\ell_2}$ become orthogonal. The asymptotic sum rate can be further simplified into

$$\bar{R}_{\text{sum}} = \frac{1}{2} \sum_k \sum_\ell \log \left(1 + \gamma M^2 c_{k,\ell}^2 p a_{k,\ell} \right). \quad (51)$$

The sum rate in (51) degrades to the summation of KL SISO transmission rates. To achieve the sum rate in (50), each UT needs to perform joint detection of the received signals. In the case of (51), each UT can separately detect the signals from each BS, which further simplifies the complexity of receive designs. The proposed transmission scheme can be applied for both static and mobile scenarios. When one UT moves from the coverage of one beam to another, the BS only needs to switch the corresponding beam to serve that UT.

B. Sum Rate Scaling Analysis

Under the per LED power constraint, we can calculate the DoF of BDMA transmission as follows.

Theorem 7: The DoF of BDMA transmission under the per LED power constraint is

$$\begin{aligned} DoF &= \lim_{M \rightarrow \infty} \frac{\frac{1}{2} \sum_k \sum_\ell \log \left(1 + \gamma M^2 c_{k,\ell}^2 p a_{k,\ell} \right)}{\log(M)} \\ &= KL. \end{aligned} \quad (52)$$

Remark 12: Similarly with the result under the total power constraint, the DoF of BDMA transmission under the per LED power constraint is also KL . This result implies that with a large number of LEDs and photodetectors, one BS can simultaneously transmits K users' signals and one UT are able to detect signals from L BSs. Consequently, the sum rate of

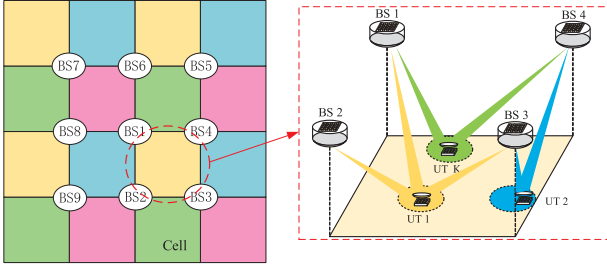


Fig. 5. Configuration of networked optical massive MIMO communication systems.

the BDMA transmission scales linearly with the number of BSs and UTs.

For comparison, consider the single-user case where a BS transmits signals to a UT without lens at both the BS and UT sides. To maximize the single user rate under the per LED power constraint, the optimal transmit covariance matrix is $\mathbf{Q} = p\mathbf{1}_{M \times M}$, and thus, the maximal single-user rate is

$$R_1^o = \frac{1}{2} \log(1 + \gamma h^2 M^2 N p). \quad (53)$$

Then, the ratio of the sum rate of BDMA transmission to the single-user rate is

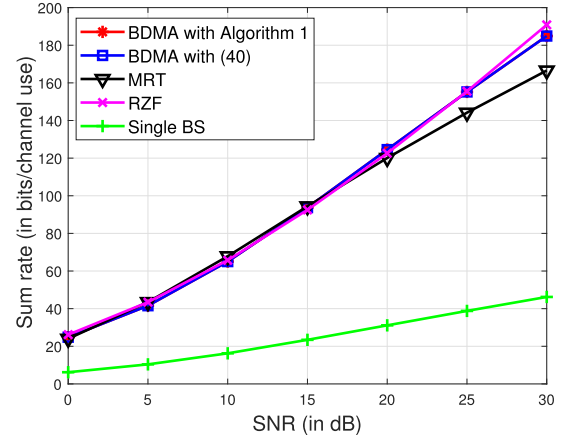
$$\begin{aligned} \lim_{M \rightarrow \infty} \frac{\bar{R}_{\text{sum}}}{R_1^o} &= \lim_{M \rightarrow \infty} \frac{\frac{1}{2} \sum_k \sum_\ell \log(1 + \gamma M^2 c_{k,\ell}^2 p a_{k,\ell})}{\frac{1}{2} \log(1 + \gamma h^2 M^2 N p)} \\ &= KL. \end{aligned} \quad (54)$$

The ratio under the per LED power constraint is different from that under the total power constraint in (47). Under the total power constraint, BDMA transmission can increase the sum rate proportionally to $2KL$, while under the per LED power constraint, the sum rate of BDMA transmission is improved by KL times. The difference is caused by the different power gains under these two power constraints.

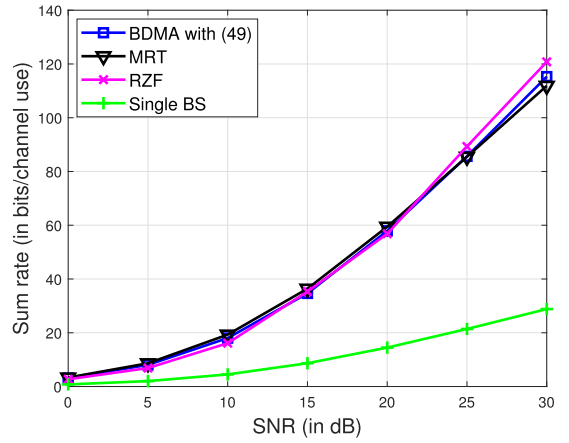
V. SIMULATION RESULTS

In this section, we demonstrate the performance of proposed networked optical massive MIMO communications with transmit and receive lenses via numerical simulations. The communication area is covered by multiple BSs, as illustrated in Fig. 5, where each small (colored) square area is a cell and covered by the adjacent four BSs, and each BS generates beams to serve four cells. Without loss of generality, we focus on the performance of one cell, and the performances of other cells are similar. We consider two typical communication scenarios. One is the micro scenario, such as a meeting room, where the size of each cell is $5\text{m} \times 5\text{m}$, and the height is 3m . In this scenario, there are four BSs, each with a transmit lens and 12×12 LEDs, simultaneously communicating with 12 UTs, each with a receive lens and 4×4 photodetectors. The other scenario is the macro scenario, for example an airport lounge or a stadium, where the size of the cell is $16\text{m} \times 16\text{m}$, and the height is 8m . In the macro scenario, four BSs, each with a transmit lens and 80×80 LEDs, serve 500 UTs.

We compare the sum rate of BDMA transmission with linear precoding based transmissions, including maximum ratio



(a)



(b)

Fig. 6. Comparison of the sum rate in the micro scenario under the (a) total and (b) per LED power constraints.

transmission (MRT) and regularized zero-forcing (RZF) transmissions. These two precoding schemes are typical and commonly used in RF wireless communications. The MRT scheme is simpler, whereas RZF can reduce the inter-user interference and provide better performance. Specifically, the transmitted signal from BS ℓ to UT k can be expressed as

$$\mathbf{x}_{k,\ell} = \mathbf{w}_{k,\ell} s_{k,\ell}, \quad (55)$$

where $s_{k,\ell}$ is the i.i.d. real symbol with unit variance, and $\mathbf{w}_{k,\ell}$ is the precoding vector. For the MRT and RZF precoding, the precoding vector $\mathbf{w}_{k,\ell}^{\text{MRT}}$ and $\mathbf{w}_{k,\ell}^{\text{RZF}}$ are given respectively by [32]

$$\mathbf{w}_{k,\ell}^{\text{MRT}} = \sqrt{\beta_{k,\ell}^{\text{MRT}}} \mathbf{g}_{k,\ell} \quad (56)$$

$$\mathbf{w}_{k,\ell}^{\text{RZF}} = \sqrt{\beta_{k,\ell}^{\text{RZF}}} (\mathbf{G}_\ell \mathbf{G}_\ell^T + \alpha \mathbf{I}_M)^{-1} \mathbf{g}_{k,\ell}, \quad (57)$$

where $\mathbf{G}_\ell = [\mathbf{g}_{1,\ell}, \mathbf{g}_{2,\ell}, \dots, \mathbf{g}_{K,\ell}]$, $\alpha > 0$ is a regularization parameter, $\beta_{k,\ell}^{\text{MRT}}$ and $\beta_{k,\ell}^{\text{RZF}}$ normalize the transmit power to satisfy the power constraint. Note that due to the matrix inversion, the complexity of the RZF precoding becomes extremely high with massive LEDs at the BS. In contrast, the BDMA scheme using the power allocation in (40) avoids the matrix inversion and the complexity scales with the number

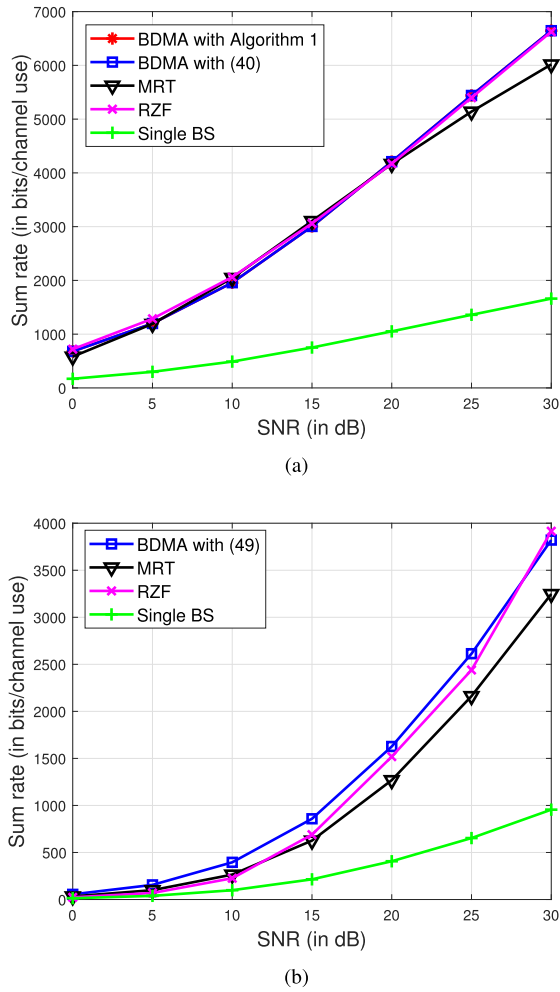


Fig. 7. Comparison of the sum rate in the macro scenario under the (a) total and (b) per LED power constraints.

of UTs, which is much smaller than that of LEDs. Thus, the BDMA scheme can significantly reduce the computational complexity.

Fig. 6 illustrates the sum rate in the micro scenario under the total and per LED power constraints. The total power under both constraints is the same, i.e., given the total power P_ℓ , the per LED power is $p = P_\ell/M$. As we consider the unit noise power, the SNR is equal to the total power P_ℓ . Under the total power constraint, the BDMA scheme based on Algorithm 1 and the BDMA scheme using the power allocation in (40) achieve almost the same sum rate. This indicates that each UT with a receiver lens can separate the signals from different BSs, and there is little interference between BSs. One can also observe that, in high SNRs, the performance of the BDMA transmission is close to that of the RZF precoding and better than that of the MRT, while the complexity of BDMA is much lower than that of RZF. Under the per LED power constraint, when the number of UTs is not large, these three transmission schemes approaches the similar performance in low and middle SNRs. Moreover, we also show the sum rate of a single BS, which is much lower than that of the networked massive MIMO system.

Fig. 7 compares the sum rate in the macro scenario under both power constraints. Under the total power constraint,

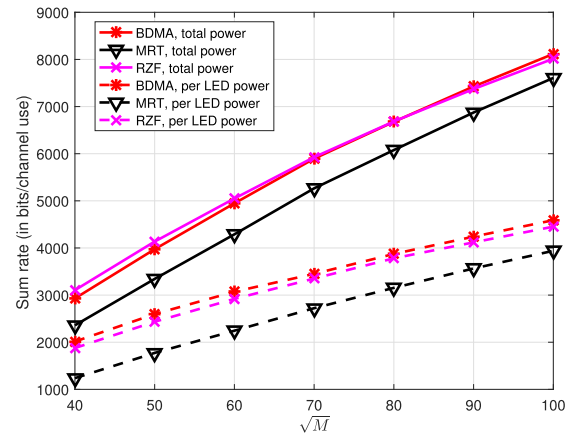


Fig. 8. Comparison of sum rate for different number of LEDs under the total power constraint.

the sum rate of BDMA transmission approaches that of RZF. Under the per LED power constraint, as the number of UTs is large, to satisfy the power constraint, the power scalar degrades the performance of linear precoding schemes. BDMA can achieve the highest sum rate, while MRT suffers about 2.5 dB performance loss at high SNR regimes. Moreover, it can be also observed that, compared with a single BS, the networked optical massive MIMO system can significantly increase the system throughput.

From Fig. 6 and Fig. 7, we can calculate the transmission rate per UT. In the micro scenario, as the number of served UTs is 12, the transmission rate per UT for 30 dB is about 15 bits per channel use under the total power constraint, while in the macro scenario with 500 UTs, the transmission rate per UT can be calculated as 13.2 bits per channel use. In contrast, the transmission rate with single BS is up to 4.5 bits per channel use in [19]. Therefore, multiple BSs can significantly improve the transmission rate per UT.

Finally, we compare the asymptotic performance of different transmission schemes as the number of LEDs increases, as illustrated in Fig. 8. Here, we consider the macro scenario with high SNR regimes (SNR is 30 dB), and increase the number of LEDs from 40×40 to 100×100 . Under both power constraints, the sum rate of BDMA can approach that of RZF with negligible performance loss, while MRT always suffers performance loss compared with RZF and BDMA. This indicates that the BDMA transmission can achieve the asymptotically optimal performance.

VI. CONCLUSION

We have investigated networked optical massive MIMO wireless communications in the beam domain with transmit and receive lenses. We provided the beam domain channel model and showed that as the transmit LEDs increases, the channel matrices from one BS to two UTs become asymptotically row orthogonal, while when the number of receivers is large, the channel matrices from two BSs to one UT become asymptotically column orthogonal. This indicates that one BS can transmit independent signals to different UTs and each UT can receive signals from different BSs. Then, we designed the transmit covariance matrices maximizing the

sum rate and proposed networked BDMA transmission under both the power constraints. In the asymptotic case, the optimal transmission strategy is that each BS utilizes one beam to transmit signals to one UT and that the transmitting beams for different UTs are non-overlapping. The optimal receiving strategy is that each UT performs the single user detection for the signals from one BS. Under the optimal transmission, the sum rate of BDMA transmission scales with the number of BSs and UTs. Simulation results presented the performance advantages and high spectrum efficiency of the networked optical massive MIMO communication system.

APPENDIX A PROOF OF THEOREM 2

Let $\mathbf{Q}_\ell = \sum_k \mathbf{Q}_{k,\ell}$. The asymptotic sum rate $\bar{R}_{\text{sum}}$ in (29) can be rewritten as

$$\begin{aligned} \bar{R}_{\text{sum}} = & \frac{1}{2} \sum_k \log \det \left(\mathbf{I} + \sum_\ell M^2 c_{k,\ell}^2 [\mathbf{Q}_\ell]_{m_{k,\ell} m_{k,\ell}} \mathbf{C}_{k,\ell} \right. \\ & \left. - (1 - \gamma) M^2 c_{k,\ell}^2 [\mathbf{Q}_{k,\ell}]_{m_{k,\ell} m_{k,\ell}} \mathbf{C}_{k,\ell} \right) \\ & - \frac{1}{2} \sum_k \log \det \left(\mathbf{I} + \sum_\ell M^2 c_{k,\ell}^2 [\mathbf{Q}_\ell]_{m_{k,\ell} m_{k,\ell}} \mathbf{C}_{k,\ell} \right. \\ & \left. - M^2 c_{k,\ell}^2 [\mathbf{Q}_{k,\ell}]_{m_{k,\ell} m_{k,\ell}} \mathbf{C}_{k,\ell} \right). \end{aligned} \quad (58)$$

As the asymptotic sum rate $\bar{R}_{\text{sum}}$ depends on the diagonal elements of $\mathbf{Q}_\ell$ and $\mathbf{Q}_{k,\ell}$, diagonal matrices $\mathbf{Q}_\ell$ and $\mathbf{Q}_{k,\ell}$ can achieve optimal performance. Let $\mathbf{Q}_{k,\ell} = \mathbf{B}_{k,\ell} \mathbf{Q}_\ell$, where $\mathbf{B}_{k,\ell}$ is an auxiliary diagonal matrix satisfying $\mathbf{B}_{k,\ell} \succeq \mathbf{0}$. Then, problem (31) is equivalent to

$$\begin{aligned} & \max_{\mathbf{Q}_\ell} \max_{\mathbf{B}_{k,\ell}} \bar{R}_{\text{sum}} \\ & \text{s.t. } \mathbf{Q}_\ell \succeq \mathbf{0}, \quad \text{tr}(\mathbf{Q}_\ell) \leq P_\ell \\ & \quad \mathbf{B}_{k,\ell} \succeq \mathbf{0}, \quad \sum_k \mathbf{B}_{k,\ell} = \mathbf{I}. \end{aligned} \quad (59)$$

For arbitrary $\mathbf{Q}_\ell$, consider the inner optimization problem on $\mathbf{B}_{k,\ell}$,

$$\begin{aligned} & \max_{\mathbf{B}_{k,\ell}} \frac{1}{2} \sum_k \log \det \left(\mathbf{I} + \sum_\ell M^2 c_{k,\ell}^2 [\mathbf{Q}_\ell]_{m_{k,\ell} m_{k,\ell}} \mathbf{C}_{k,\ell} \right. \\ & \quad \left. - (1 - \gamma) M^2 c_{k,\ell}^2 [\mathbf{B}_{k,\ell} \mathbf{Q}_\ell]_{m_{k,\ell} m_{k,\ell}} \mathbf{C}_{k,\ell} \right) \\ & \quad - \frac{1}{2} \sum_k \log \det \left(\mathbf{I} + \sum_\ell M^2 c_{k,\ell}^2 [\mathbf{Q}_\ell]_{m_{k,\ell} m_{k,\ell}} \mathbf{C}_{k,\ell} \right. \\ & \quad \left. - M^2 c_{k,\ell}^2 [\mathbf{B}_{k,\ell} \mathbf{Q}_\ell]_{m_{k,\ell} m_{k,\ell}} \mathbf{C}_{k,\ell} \right) \\ & \text{s.t. } \mathbf{B}_{k,\ell} \succeq \mathbf{0}, \quad \sum_k \mathbf{B}_{k,\ell} = \mathbf{I}. \end{aligned} \quad (60)$$

Similarly to Lemma 2 in [33], $\bar{R}_{\text{sum}}$ is convex on $[\mathbf{B}_{k,\ell}]_{m_{k,\ell} m_{k,\ell}}$. To maximize the asymptotic sum rate $\bar{R}_{\text{sum}}$,

$[\mathbf{B}_{k,\ell}]_{m_{k,\ell} m_{k,\ell}}$ should satisfy

$$[\mathbf{B}_{k,\ell}]_{m_{k,\ell} m_{k,\ell}} [\mathbf{B}_{k',\ell}]_{m_{k',\ell} m_{k',\ell}} = 0. \quad (61)$$

Thus, the optimal $\mathbf{Q}_{k,\ell}$ satisfies

$$\mathbf{Q}_{k,\ell} \mathbf{Q}_{k',\ell} = \mathbf{0}, \quad k \neq k'. \quad (62)$$

In addition, at most one diagonal element of $\mathbf{Q}_{k,\ell}$ is non-zero. As $\mathbf{Q}_{k,\ell}$ is a semi-definite positive matrix, $\mathbf{Q}_{k,\ell}$ can only be a diagonal matrix. This completes the proof. ■

APPENDIX B PROOF OF THEOREM 3

Let $\mu_{k,\ell}$ be the Lagrange multiplier associated with the inequality constraint $q_{k,\ell} \geq 0$ and η_ℓ be the Lagrange multiplier associated with the inequality constraint $\sum_k q_{k,\ell} \leq P_\ell$. We have the cost function of the optimal design as

$$\begin{aligned} \mathcal{L} = & \frac{1}{2} \sum_k \log \det \left(\mathbf{I} + \sum_\ell q_{k,\ell} \mathbf{h}_{k,\ell} \mathbf{h}_{k,\ell}^T \right) \\ & + \sum_{k,\ell} q_{k,\ell} \mu_{k,\ell} + \eta_\ell \left(P_\ell - \sum_k q_{k,\ell} \right). \end{aligned} \quad (63)$$

The Karush-Kuhn-Tucker (KKT) conditions for optimum $q_{k,\ell}$, $\mu_{k,\ell}$ and η_ℓ can be expressed as

$$\begin{aligned} & \mathbf{h}_{k,\ell}^T \left(\mathbf{I} + \sum_j q_{k,j} \mathbf{h}_{k,j} \mathbf{h}_{k,j}^T \right)^{-1} \mathbf{h}_{k,\ell} + \mu_{k,\ell} - \eta_\ell = 0, \\ & q_{k,\ell} \mu_{k,\ell} = 0, \quad \eta_\ell \left(P_\ell - \sum_k q_{k,\ell} \right) = 0, \quad \mu_{k,\ell}, \eta_\ell \geq 0. \end{aligned} \quad (64)$$

Employing matrix inverse lemma, we have

$$\begin{aligned} & \mathbf{h}_{k,\ell}^T \left(\mathbf{I} + \sum_j q_{k,j} \mathbf{h}_{k,j} \mathbf{h}_{k,j}^T \right)^{-1} \mathbf{h}_{k,\ell} \\ & = \frac{\mathbf{h}_{k,\ell}^T \left(\mathbf{I} + \sum_{j \neq \ell} q_{k,j} \mathbf{h}_{k,j} \mathbf{h}_{k,j}^T \right)^{-1} \mathbf{h}_{k,\ell}}{1 + q_{k,\ell} \mathbf{h}_{k,\ell}^T \left(\mathbf{I} + \sum_{j \neq \ell} q_{k,j} \mathbf{h}_{k,j} \mathbf{h}_{k,j}^T \right)^{-1} \mathbf{h}_{k,\ell}}. \end{aligned} \quad (65)$$

When $q_{k,\ell} > 0$, we can have $\mu_{k,\ell} = 0$ and

$$q_{k,\ell} = \frac{1}{\eta_\ell} - \frac{1}{\mathbf{h}_{k,\ell}^T \left(\mathbf{I} + \sum_{j \neq \ell} q_{k,j} \mathbf{h}_{k,j} \mathbf{h}_{k,j}^T \right)^{-1} \mathbf{h}_{k,\ell}}, \quad (66)$$

where η_ℓ satisfies the power constraint $\sum_k q_{k,\ell} = P_\ell$. This completes the proof. ■

APPENDIX C PROOF OF THEOREM 6

Similarly with the proof of Theorem 2, we can prove that the optimal solution of problem (48) satisfies

$$\mathbf{Q}_{k_1,\ell} \mathbf{Q}_{k_2,\ell} = \mathbf{0}, \quad k_1 \neq k_2. \quad (67)$$

Under this condition, the asymptotic sum rate can be expressed as

$$\bar{R}_{\text{sum}} = \frac{1}{2} \sum_k \log \det \left(\mathbf{I} + \gamma \sum_\ell M^2 c_{k,\ell}^2 [\mathbf{Q}_{k,\ell}]_{m_{k,\ell} m_{k,\ell}} \mathbf{C}_{k,\ell} \right). \quad (68)$$

To maximize the asymptotic sum rate, the optimal transmit covariance matrix $\mathbf{Q}_{k,\ell}$ is

$$[\mathbf{Q}_{k,\ell}]_{m_{k,\ell}m_{k,\ell}} = p. \quad (69)$$

This completes the proof. ■

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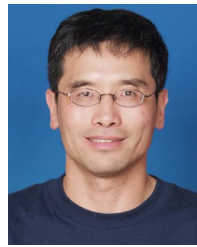


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